Scaling Surface Haptics: Design and Applications of a Low-Cost, Single-Terminal Flexible Tactile Tape

JIEXIN DING, HONGYU MAO, SELAH KEY

Flexible tactile sensors play a critical role in enabling interactive environments, smart robotics, and wearable devices. However, scaling grid-based tactile sensors to long, tape-like form factors remains challenging due to side-routing complexity and high wire density. This paper presents a 1m-long, 3cm-wide flexible tactile sensor tape that eliminates complex side-routing by placing all connection pins at a single terminal end. Our design employs a matrix-sensing layout leveraging dynamic RC delay differentiation to distinguish multi-touch positions across 64 intersections (4×16 resolution). Built with low-cost, accessible materials including conductive fabric, Velostat, and foam isolation layers, the sensor tape can be modularly segmented and seamlessly integrated into long-form surfaces. We demonstrate its utility through three applications: an interactive plush toy, a smart racket handle, and a tactile music tape. Preliminary evaluations indicate that our approach provides a scalable and cost-effective solution for large-scale surface haptics.

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1 INTRODUCTION

The demand for large-scale, flexible tactile interfaces has grown rapidly with the rise of ubiquitous computing, soft robotics, and smart environments. Transforming everyday linear or long-form structures, such as architectural handrails, robotic limbs, or sports equipment handles, into pressure-sensitive interfaces could unlock novel modalities for human-computer interaction, sports analytics, and accessibility.

However, scaling traditional grid-based tactile arrays to extended lengths (e.g., 1 meter or more) introduces significant hardware and structural challenges. An $N \times M$ matrix sensor conventionally requires $N + M$ dedicated traces routed along the substrate edges to reach the readout circuit. In a long, narrow form factor, this side-routing consumes the limited tape width and grows with sensor length, restricting flexibility, inflating manufacturing cost, and ultimately capping the achievable sensing resolution. As a result, existing long-form tactile interfaces often resort to numerous external wires or rigid components, rendering them difficult to wrap around non-planar or dynamic surfaces.

To address these limitations, we introduce a flexible tactile sensor tape, 1 meter long and 3 cm wide, that aggregates all electrical connections at a single terminal end. Two design elements make this possible. First, a single-sheet foldable layout routes every column and row trace toward one end of the tape, eliminating edge routing entirely. Second, passive RC delay coding compresses the readout interface. Inline capacitors of four distinct orders of magnitude (22 nF to 22 μF) give each row group a distinct decay duration, so four shared GPIO pins suffice to read all sixteen rows. Because extending the tape adds row groups that reuse the same shared pins, the readout pin count does not grow with tape length.

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To ensure low cost and scalability, the tape is fabricated using accessible materials, including conductive fabric electrodes, a piezoresistive Velostat layer for pressure sensing, and laser-cut foam spacers to prevent crosstalk and signal spreading. In this work, we detail our circuit and fabrication design iterations, demonstrate the optimization of capacitive groups to maximize position resolution, and present three functional application prototypes.

2 RELATED WORK

2.1 Large-Scale and Long-Form Tactile Interfaces

Tactile sensing has been scaled well beyond the fingertip: modular floorspaces [11], conductive inkjet surfaces [2], room-scale electrode wallpaper [17], a 548-sensor piezoresistive glove [13], e-textiles [8, 9], and large-area robotic e-skins [5]. These systems target planar geometries, however, with row–column interconnects fanning out along multiple substrate edges, a topology that fails when one dimension grows much longer than the other. Among long, narrow form factors, I/O Braid senses gestures along textile cords but does not localize pressure [7], and cuttable printed sheets target physical reconfigurability [6]. Most closely related, SensorTape embeds a microcontroller and sensors on every node, so each segment becomes an active, costly, and stiff module [1]. In contrast, our sensing surface is entirely passive, with all active electronics consolidated at a single terminal end.

2.2 Wiring Reduction and Analog Multiplexing in Sensor Matrices

Reducing matrix wiring ($N+M$ traces for an $N\times M$ grid) has been approached through tomography, which recovers touch from boundary electrodes alone at the cost of heavy reconstruction and limited resolution [15, 16], and through signal-domain separation: swept-frequency profiling [12], time domain reflectometry [14], and frequency- or code-division driving [4]. Our RC delay differentiation belongs to this time-domain family but encodes channel identity passively through inline capacitors, extending the tradition of accessible prototyping with commodity materials [3, 10, 13] toward single-terminal, long-form sensing.

3 SYSTEM

3.1 Circuit Design and RC Delay Differentiation

To eliminate complex side-routing and minimize wire density on the 1-meter tape, we designed a scanning matrix with a one-end pinning routing mechanism that aggregates all connection interfaces at a single terminal end. The finalized hardware architecture consists of 4 vertical columns (Cols) governed by a multiplexer (MUX) and 16 horizontal rows (Rows) split into 4 distinct row groups (Groups 1–4), establishing a 4×16 tactile matrix with 64 localized intersections. Within this matrix, inline surface-mount capacitors are utilized uniquely to differentiate the four row groups. Each group contains 4 row traces hardwired to a specific, unique order of magnitude of capacitance ($C_1 = 22$ nF, $C_2 = 220$ nF, $C_3 = 2.2$ μF, $C_4 = 22$ μF). The specific row index within any group is directly read via 4 dedicated ESP32 GPIO pins (Row_{pins} [0–3]), which are shared symmetrically across the groups (e.g., Row 1 of every group maps to GPIO 25).

The spatial localization of a single touch event operates on a hybrid multiplexing and analog RC delay differentiation principle. The MUX sequentially drives voltage pulses into each column, while the ESP32 samples the shared row pins. When a user presses an intersection, the piezoresistive Velostat layer undergoes mechanical deformation, causing its contact resistance (R_{velostat}) to drop dynamically. This sudden conductive pathway closes the circuit, generating a voltage step from 0V to a positive peak value ($V_{\text{initial}} > 1.0$V) at the target ESP32 read pin due to the voltage divider established between R_{velostat} and a 9 kΩ pull-down resistor (R_1). Immediately following this voltage step, the corresponding inline

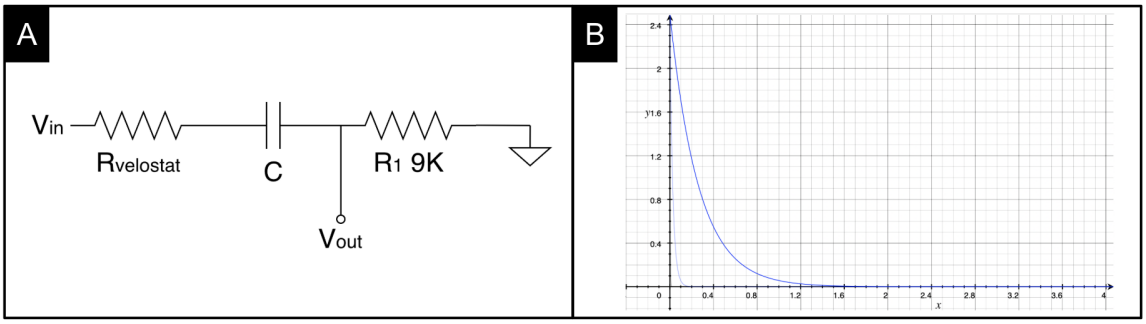


Fig. 1. Core sensing circuit and signal differentiation. (A) Simplified RC equivalent circuit model formed at a pressed touch intersection, establishing a dynamic voltage divider network. (B) Exponential voltage decay profiles across the time domain (V_{out} over t), showing distinct time constants (τ) engineered to uniquely resolve capacitive row groups.

capacitor begins to charge, causing the pin voltage to decay exponentially towards ground (0V) until the capacitor is fully charged and acts as an open circuit. The ESP32 high-frequency ADC continuously polls the pin to capture the exact time duration (Δt) required for the voltage to decay past a low threshold (0.4V). Because each row group features a vastly different capacitance value, their resulting RC time constants ($\tau = R \times C$) produce strictly segregated delay thresholds ($\sim 10^2$ μs to 10^5 μs). Thus, the system resolves coordinates by identifying the activated column from the active MUX channel, the row group from the time-domain decay duration, and the specific row line from the triggered GPIO pin.

3.2 Mitigation of Structural Crosstalk and Parasitic Capacitance

Scaling this time-domain multiplexing to a 1-meter flexible substrate exposed two interconnected hardware challenges: inter-electrode crosstalk and unwanted parallel capacitance. In a continuous sheet architecture, parasitic leakage across adjacent traces causes capacitors from unpressed groups to parallelize or serialize, altering the baseline network capacitance and completely corrupting the distinct RC delay profiles. We resolved these challenges through structural isolation:

- Row and Group-Wise Isolation: The piezoresistive Velostat layer is mechanically sliced into discrete, separated horizontal strips. This physically terminates vertical current leakage, preventing cross-group capacitor serialization and eliminating row-to-row crosstalk.
- Column-Wise Isolation: A laser-cut foam spacer layer is introduced between the Velostat strips and the vertical conductive fabric columns. The foam layer features a matrix of 64 micro-cavities precisely aligned with the electrode intersections, preventing unwanted capacitive coupling between neighboring columns.

Beyond suppressing crosstalk, the compressed foam spacer introduces a mechanical spring-back effect. Upon touch release, the foam immediately pushes the conductive layers apart, driving the resting output to absolute zero. Without this mechanical restoring force, the Velostat maintains a large but finite residual baseline conductivity even in its unpressed state. Because R_{velostat} is inherently unstable and pressure-dependent, an unpressed node with a large capacitance (e.g., Group 4) could generate a decay duration identical to an actively pressed node with a small capacitance (e.g., Group 3). The combination of structural slicing and foam-gated mechanical feedback ensures that unpressed nodes

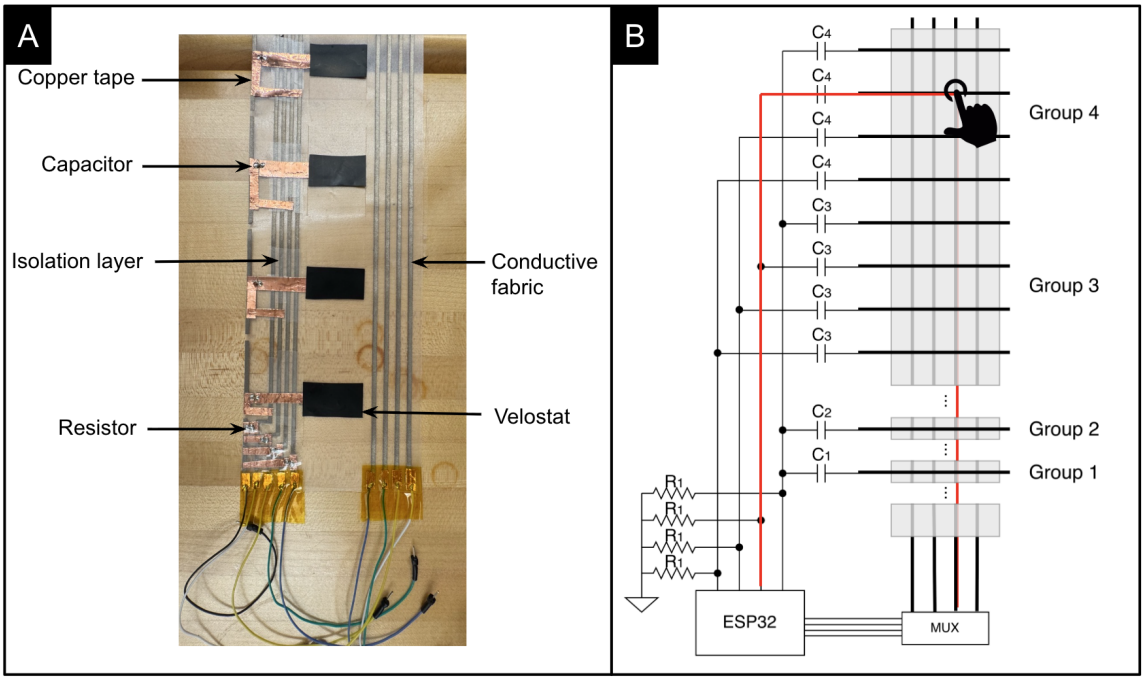


Fig. 2. Hardware prototype and full system schematic. (A) Annotated photograph of the flexible tactile tape layout demonstrating terminal-end component integration and layer stacking. (B) Full circuit diagram illustrating the matrix routing network, where a shared 4-channel ESP32 readout pinout interfaces with four capacitive row groups sequentially driven by MUX column-switching.

remain completely decoupled, maintaining a robust, highly repeatable delay differentiation across the entire 1-meter span.

3.3 Fabrication

The sensor integrates a matrix sensing structure, wiring traces, and discrete capacitors into a single continuous layout that is patterned flat and then folded into the final stack, avoiding the bulky side routing of conventional long-form matrices. The folded assembly comprises three functional layers: vertical electrodes, horizontal electrodes, and a wiring layer carrying the routing traces and capacitor pads, with Velostat placed between the electrode layers as the pressure-sensitive material. This arrangement keeps the three layers spatially separated during patterning while vertically aligned after folding. The sensing area is tiled into repeated 4×4 electrode units along the length of the tape, forming a 1 m, 16×4 sensor in which each group is assigned a distinct capacitor value, and all electrical connections route toward a single terminal end.

We fabricated the tape from laser-cut conductive fabric laminated onto an adhesive backing, a strategy chosen for rapid prototyping and scalable patterning. Two cutting settings were used: the first cut through the conductive fabric alone to define the electrodes, wiring traces, and capacitor pads, while the second cut through both fabric and adhesive to define the sensor boundary and fold lines. After removing the unwanted fabric, capacitors were attached to the connection pads, Velostat was inserted between the electrode layers, and a foam spacer was added to isolate

non-sensing regions. The substrate was then folded along the fold lines into the final stack, yielding a 1 m flexible tape with a compact geometry, distributed sensing area, and single-terminal interface.

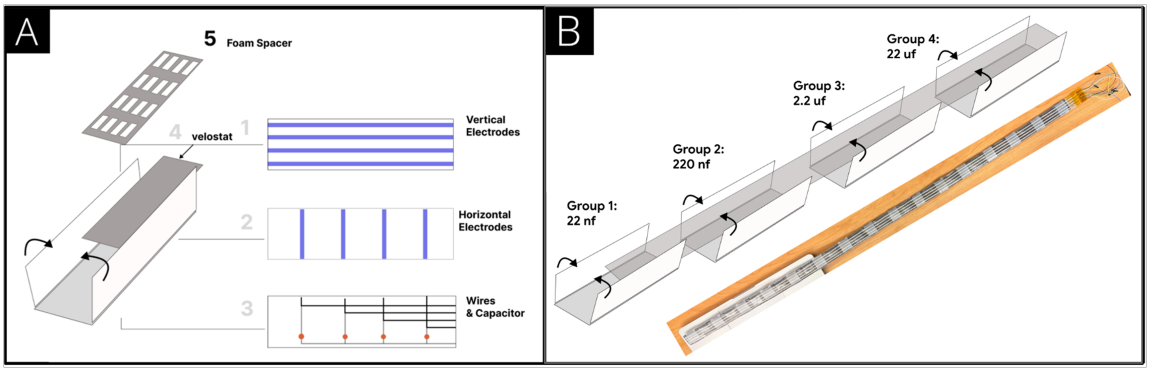


Fig. 3. (A) Schematic illustration of the three-fold sensor architecture. The sensor integrates vertical electrodes, horizontal electrodes, wiring traces, and capacitor connection pads into a single foldable layout. Velostat is inserted between the electrode layers as the pressure-sensitive material, while a foam spacer is added to improve signal isolation, reduce pressure spreading, and suppress crosstalk between neighboring sensing regions. (B) Long-form sensor design with repeated sensing groups along the tape. Each group contains a 4 × 4 electrode matrix and is assigned a distinct capacitor value to generate a different RC delay range, enabling segment-wise touch localization along the 1 m tape. The folded structure allows distributed sensing units to be integrated along the full length while routing all electrical connections to one terminal end.

4 RESULT

We evaluated the performance of the 1-meter flexible tactile sensor tape with a focus on crosstalk suppression, multi-touch detection capability, and the temporal characteristics of the RC delay profiles.

4.1 Crosstalk Suppression

Owing to the structural isolation framework, specifically the horizontal slicing of the Velostat layer and the implementation of the micro-cavity perforated foam spacer, inter-electrode crosstalk was effectively eliminated. The mechanical spring-back effect of the foam spacer successfully isolated inactive intersections, preventing parasitic parallel or serial connections between adjacent capacitive networks. Consequently, when a single or multiple non-adjacent nodes were pressed, no signal spreading or ghost touches were observed, and the baseline delay values of unpressed nodes remained strictly at zero.

4.2 Multi-Touch Detection Capabilities and Limitations

The system demonstrated robust multi-touch detection across the 4×16 matrix, successfully mapping simultaneous touch points at arbitrary coordinates with one specific hardware-bound exception. Because rows across different capacitive groups share the same ESP32 readout GPIO pins, a geometric ambiguity arises if a user simultaneously presses the exact same row index (e.g., Row 1) across different row groups (e.g., Group 1 and Group 4) within the same driven column. In this specific edge case, the parallel connection of the two capacitors alters the equivalent RC time constant on that shared pin, preventing the decoding algorithm from separating the individual nodes. However, outside

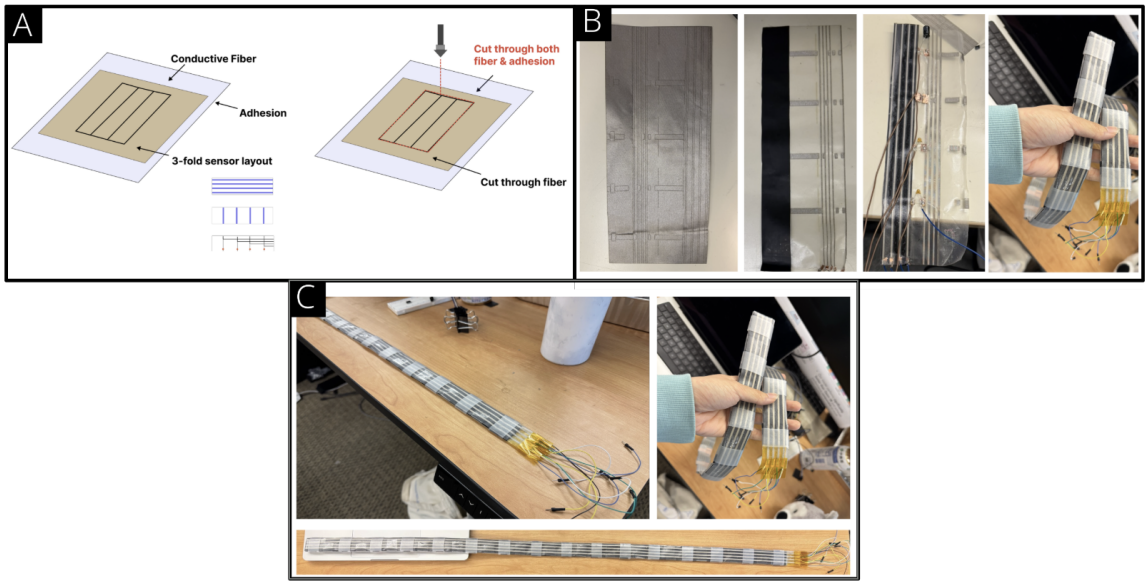


Fig. 4. (A) Schematic illustration of the laser-cut fabrication strategy. Conductive fabric was laminated onto an adhesive substrate and patterned into a three-fold sensor layout. Two laser-cutting settings were used: one selectively cut through the conductive fabric to define the electrode and wiring patterns, while the other cut through both the conductive fabric and adhesive layer to define the sensor boundary and folding regions. (B) Fabrication and assembly process of an early prototype, including the patterned conductive fabric, integration of Velostat and sensing layers, attachment of capacitors and wiring, and folding into a tape-like form factor. (C) Final 1 m flexible sensor tape prototype. The fabricated sensor preserves a narrow and flexible tape geometry while integrating distributed sensing units along the full length and routing all electrical connections to one terminal end.

of this shared-pin, intra-column alignment condition, the sensor tape reliably isolates and visualizes multi-touch profiles across the entire 1-meter surface.

4.3 Temporal Response and Delay Analysis

The time-domain multiplexing latency was heavily dictated by the capacitance order of magnitude assigned to each row group:

- Groups 1–3 (22 nF to 2.2 μF): Exhibited rapid exponential decay durations ($\sim 10^2$ μs to 10^4 μs), allowing for near-instantaneous scanning.
- Group 4 (22 μF): Due to the high capacitance required for distinctive group profiling, the largest row group exhibited a typical scanning delay of approximately 100 ms under normal operating conditions. Under extreme contact resistance scenarios (e.g., light touch inputs), the maximum discharge and timeout latency peaked at approximately 400 ms.

While the latency of Group 4 bounds the overall refreshing rate of the extended tape, preliminary user testing in the interactive toy and music tape applications confirmed that the response speed remains highly acceptable for large-scale surface interactions and posture monitoring.

5 APPLICATION

We deployed the tape in three interactive prototypes that exercise its ability to conform to curved, dynamic, everyday surfaces while maintaining robust spatial sensing.

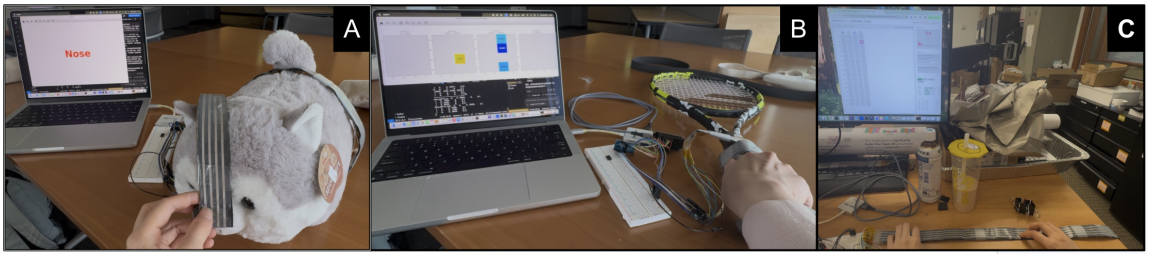


Fig. 5. (A) Interactive plush toy application for early childhood education. Contact at designated coordinates (e.g., the nose area) triggers localized semantic vocabulary feedback on the connected interface. (B) Smart racket handle application for sports analytics. Real-time hand grasp profiles and multi-touch pressure coordinates are captured and visualized as heatmaps during tactical movements. (C) Tactile music tape demonstration. The linear sensor array acts as an accessible instrument surface, mapping sequential matrix inputs to real-time audio parameters and musical notes.

5.1 Interactive Plush Toy for Early Childhood Education

By wrapping the flexible tactile tape around a standard stuffed animal, we transformed a passive toy into an interactive, touch-sensitive agent. In this prototype, the 64 discrete sensing intersections of the matrix are mapped to specific physiological regions of the toy (e.g., nose, head, body, leg). When a child presses a particular area, the system successfully resolves the coordinates through our RC delay differentiation mechanism and triggers localized voice feedback corresponding to that body part. This setup serves as an engaging, low-cost tangible user interface designed for early childhood language learning and cognitive education. By binding physical tactile exploration with real-time auditory responses, the interactive plush toy gamifies vocabulary acquisition and structural learning in an accessible form factor.

5.2 Smart Racket Handle for Sports Analytics

To showcase the tape's integration into high-curvature, sports-oriented surfaces, we seamlessly wrapped the tactile tape around the handle grip of a tennis racket. Due to the single-terminal layout, all wiring was cleanly aggregated at the base of the handle, completely avoiding bulky side-traces that would disrupt the user's natural grip. As an athlete plays, the sensor matrix captures real-time dynamic grip pressure distributions and contact locations during various swings (e.g., forehands, backhands, and smashes). This application demonstrates the tape's potential in sports analytics and physical training, providing coaches and players with precise, data-driven insights into hand ergonomics and posture monitoring without compromising the form factor of the equipment.

5.3 Tactile Music Tape

The third application explores the intersection of tactile sensing and ubiquitous audio control. We configured the 1-meter sensor tape as a lightweight, flexible playable surface in which different sections of the tape are mapped to distinct musical notes. By applying continuous pressure along the tape, a user can trigger notes and modulate sound effects in real time, turning an ordinary strip of material into an expressive instrument.

More broadly, mapping each sensing location to a distinct voice or audio cue demonstrates the potential of the tape as a surface for interactive auditory feedback. Because it is thin, conformable, and inexpensive, this capability can be embedded into a wide range of everyday surfaces, from on-body control interfaces and furniture to hospital bed rails that let patients with limited mobility signal caregivers through simple touch.

6 DISCUSSION

6.1 Design Trade-offs and Limitations

Our central design decision, encoding row-group identity through order-of-magnitude capacitance differences, trades temporal performance for spatial scalability. Because the four groups must remain strictly separable in time even under variable contact resistance, the largest capacitance (22 μF) imposes a scanning delay of roughly 100 ms, with worst-case timeouts near 400 ms under light touch. This bounds the refresh rate and makes the current design better suited to posture monitoring, grip profiling, and discrete triggering than to latency-critical gestures such as fast swipes. However, the trade-off is tunable. Tightening the capacitance ratios, raising the decay threshold, fitting the decay curve rather than waiting for a fixed crossing, or sampling the row pins in parallel could each shorten worst-case latency without redesigning the tape.

Several limitations qualify our results. The shared-pin architecture cannot separate simultaneous presses on the same row index across different groups within one driven column, a geometrically rare case that future per-group sense resistors or learned classification of composite decay signatures could resolve. Our evaluation also focused on functional characterization rather than quantitative benchmarks; localization accuracy, pressure linearity, hysteresis, and long-term durability under repeated folding remain to be measured. Finally, the prototype reports coarse intensity but is not calibrated for absolute force, and our application studies were preliminary demonstrations rather than formal user evaluations.

6.2 Scalability and Future Directions

Because group identity is encoded passively, the tape can in principle be extended beyond one meter by introducing further capacitance decades or a second discriminating dimension (e.g., distinct pull-down resistances per group), without altering the readout pin count. The foldable single-sheet fabrication lends itself to roll-to-roll production and to user-driven customization, where segments are cut, terminated, and re-characterized in the spirit of cuttable sensors [1, 6]. The decay signals themselves may also carry richer information, and learning-based decoding could recover contact attributes such as area or material from the same single-terminal signal. Integrating wireless readout and a battery at the terminal end would finally make the tape self-contained for deployment on handrails, straps, furniture, and garments.

7 CONCLUSION

We presented a 1-meter flexible tactile sensor tape that eliminates the side-routing bottleneck of long-form matrix sensors by consolidating all connections at a single terminal end. Passive RC delay differentiation across four capacitive row groups resolves 64 intersections with a readout pin count independent of tape length, using only low-cost materials. Three prototypes (a plush toy, a racket handle, and a tactile music tape) demonstrate the tape's ability to turn everyday long, narrow surfaces into expressive tactile interfaces.

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